Line Integral

A continuous vector function $F(R)$ which is define at each point of curve C in space.

$\therefore$ the total work done by F during the displacement from A to B is given by the line integral ;

$$\int_A^B F \cdot dR$$

here,

$$R = xI + yJ + zK$$

$$\partial R = \partial x\, I + \partial y\, J + \partial z\, K$$

Note:-

Multiple value varies then

take $y = t$ form or $x = t$ or $z = t$

both same.

t varies from range of $(x_1 - x_2)$ or $(y_1 - y_2)$ or $(z_1 - z_2)$ which is same.

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Q.1. If $F = 3xy\,I - y^2 J$, Evaluate $\int F \cdot dR$, where C is the curve in the xy-plane $y = 2x^2$ from (0,0) to (1,2).

⇒ Given;

$$F = 3xy\,I - y^2 J$$

Now,

$$R = xI + yJ + zK$$
$$\partial R = \partial x\,I + \partial y\,J + \partial z\,K$$

$$F \cdot dR = (3xy\,I - y^2 J) \cdot (dx\,I + dy\,J + dz\,K)$$

$$F \cdot dR = 3xy\,dx - y^2 dy \quad \text{—— (1)}$$

On the curve $y = 2x^2$

$$dy = 4x\,dx \quad \text{——}$$

putting value of y and dy in eq-(1)

$$F \cdot dR = 3x(2x^2)\,dx - (2x^2)^2 \cdot 4x\,dx$$

$$F \cdot dR = 6x^3 dx - 16x^5 dx$$

Integrate both side

$$\int_C F \cdot dR = \int 6x^3 dx - 16x^5 dx$$

here, x goes from 0 to 1

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$$= \int_{x=0}^{1} (6x^3 - 16x^5)\,dx$$

$$= \left[\frac{6x^4}{4}\right]_0^1 - \frac{16}{6}\left[x^6\right]_0^1$$

$$= \frac{6}{4} - \frac{16}{6}$$

$$= \frac{3}{2} - \frac{8}{3}$$

$$= \frac{9-16}{6}$$

$$= -7/6$$

Q.2. A vector field is given by $F = \sin y\, I + x(1+\cos y)J$. Evaluate the line integral over a circular path given by $x^2+y^2=a^2$, $z=0$.

⇒ Given;

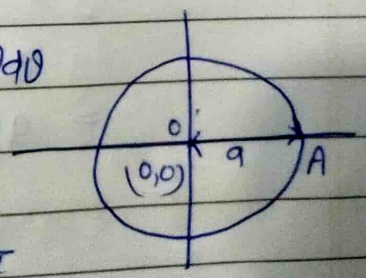
$$F = \sin y\, I + x(1+\cos y)J$$

$$dR = dx\, I + dy\, J + dz\, K$$

Line integral over a circular path

Substitute $x = a\cos\theta \Rightarrow dx = -a\sin\theta\, d\theta$

$y = a\sin\theta \quad dy = a\cos\theta\, d\theta$



θ varies from 0 to 2π

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Page No.: _____
Date: ___/___/_____

$$\int_C F \cdot dR = \int_0^{2\pi} (\sin y\, I + x(1-\cos y)\, J) \cdot (dx\, I + dy\, J + dz\, K)$$

$$= \int_0^{2\pi} \sin y\, dx + x(1-\cos y)\, dy$$

$$= \int_0^{2\pi} \sin(a\sin\theta)(-a\sin\theta)\, d\theta + a\cos\theta(1-\cos a\sin\theta)\, a\cos\theta\, d\theta$$

$$= \int_0^{2\pi} \underbrace{-a\sin\theta \cdot \sin(a\sin\theta)}\, d\theta + a^2\cos^2\theta\,(1-\cos(a\sin\theta))\, d\theta$$

~~here,~~ $\cancel{f = a\sin\theta}$
$\cancel{dz = a\cos\theta\, d\theta}$

$$= \int_0^{2\pi} \underbrace{-a\sin\theta \cdot \sin(a\sin\theta)\, d\theta}_{\text{odd function} \to \int = 0} + \int_0^{2\pi} a^2\cos^2\theta\, d\theta$$

$$- \int_0^{2\pi} a^2\cos^2\theta\cos(a\sin\theta)\, d\theta$$

$$= \int_0^{2\pi} a^2\left[\frac{1+\cos 2\theta}{2}\right] d\theta - a^2 \int_0^{2\pi} \underbrace{\cos^2\theta\cos(a\sin\theta)\, d\theta}_{\text{odd function}}$$

$\int f_x\, dx = 0$

$$= a^2 \int_0^{2\pi} \left(\frac{1}{2} + \frac{\cos 2\theta}{2}\right) d\theta$$

$$= a^2\left[\left(\frac{\theta}{2}\right]_0^{2\pi} + \left[\frac{\sin 2\theta}{4}\right]_0^{2\pi}\right]$$

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$$= a^2\left[\frac{2\pi}{2} + 0\right]$$

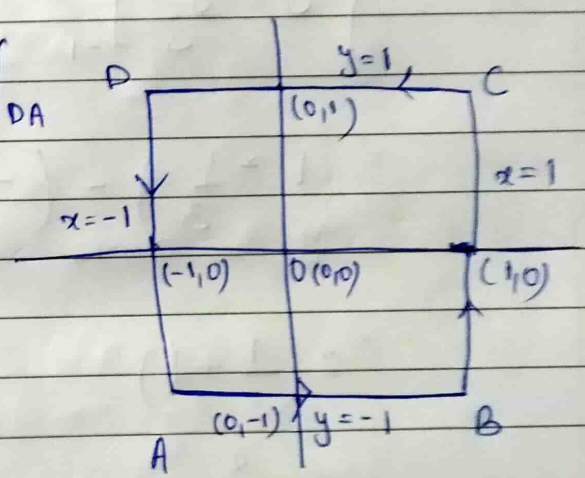
$$= \pi a^2 \text{ u.}$$

Q.2. Evaluate the line integral $\int_C (x^2+xy)\,dx + (x^2+y^2)\,dy$ where, C is the square formed by the lines $y = \pm 1$ and $x = \pm 1$.

=> Given;

$$\int_C (x^2+xy)\,dx + (x^2+y^2)\,dy$$

$$\int_C (\;) = \int_{AB} + \int_{BC} + \int_{CD} + \int_{DA}$$



On AB;

$y = -1$

$dy = 0$

$$= \int_{-1}^{1} (x^2 + x(-1))\,dx + (x^2 + (-1)^2)\cdot 0$$

$$= \int_{-1}^{1} (x^2 - x)\,dx$$

$$= \left[\frac{x^3}{3}\right]_{-1}^{1} - \left[\frac{x^2}{2}\right]_{-1}^{1}$$

$$= \frac{1}{3} + \frac{1}{3} = 2/3$$

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Page No.: _____
Date: ___/___/____

On BC;

$x = 1$

$dx = 0$

$$\int_{-1}^{1} (1^2 + y^2)\, dy$$

$$= \int_{-1}^{1} (1 + y^2)\, dy$$

$$= \left[y + \frac{y^3}{3} \right]_{-1}^{1}$$

$$= \left(1 + \frac{1}{3}\right) - \left(-1 - \frac{1}{3}\right)$$

$$= 2\left(\frac{1}{3} + 1\right)$$

$$= \frac{8}{3}$$

On CD;

$y = 1$

$dy = 0$

$$\int_{1}^{-1} (x^2 + x(1))\, dx$$

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Page No.:
Date: / /

$$\int_{1}^{-1} (x^2 + x)\, dx$$

$$= \left[\frac{x^3}{3}\right]_{1}^{-1} + \left[\frac{x^2}{2}\right]_{1}^{-1}$$

$$= \left[\frac{-1}{3} - \frac{1}{3}\right] + \left(\frac{1}{2} - \frac{1}{2}\right]$$

$$= -\frac{2}{3}$$

On DA;

$x = -1$
$dx = 0$

$$\int_{1}^{-1} ((-1)^2 + y^2)\, dy$$

$$\int_{1}^{-1} (1 + y^2)\, dy$$

$$[y]_{1}^{-1} + \left[\frac{y^3}{3}\right]_{1}^{-1}$$

$$[-1 - 1] + [-1/3 - 1/3]$$

$$= -8/3$$

$$\text{TOTAL} = \int_{AB} () + \int_{BC} () + \int_{CD} () + \int_{DA} ()$$

$$= 2/3 + 8/3 - 2/3 - 8/3 = 0$$

Akash

Page No.:
Date: / /

Q. 4 If $F = (5xy - 6x^2)\,I + (2y - 4x)\,J$, evaluate $\int_C F \cdot dR$ along the curve C in the xy-plane, $y = x^3$ from the point $(1,1)$ to $(2,8)$.

=> Given;

$$F = (5xy - 6x^2)\,I + (2y - 4x)\,J$$

$$\int_C F \cdot dR = \int_C (5xy - 6x^2)\,I + (2y - 4x)\,J \cdot (I\,dx + J\,dy + K\,dz)$$

$$= \int_C (5xy - 6x^2)\,dx + (2y - 4x)\,dy$$

Along the curve C in the xy-plane

$$y = x^3$$
$$dy = 3x^2\,dx$$

x varies ~~fom~~ from 1 to 2

$$\int_1^2 (5x(x^3) - 6x^2)\,dx + (2(x^3) - 4x) \cdot 3x^2\,dx$$

$$\int_1^2 (5x^4 - 6x^2)\,dx + (6x^5 - 12x^3)\,dx$$

$$= \left[x^5 - 2x^3\right]_1^2 + \left[x^6 - 3x^4\right]_1^2$$

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$x_2 - x_1$

$y = x +$

$(1,0)$ $\quad$ $(0,1)$

$y - 0 = x$

$y = x$

$$= \left[\left(2^5 - 2(2)^3\right) - \left((1)^2 - 2(1)^3\right) \right] + \left[\left((2)^6 - 3(2)^4\right) - \left((1)^6 - 3(1)^4\right) \right]$$

$$= [32 - 16] - [1 - 2] + [64 - 48] - [1 - 3]$$

$$= 16 + 1 + 16 + 2$$

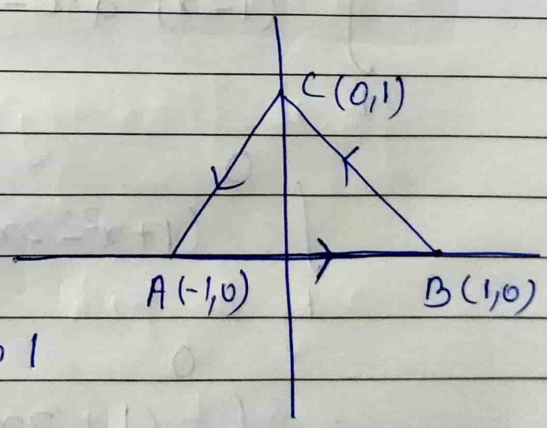
$$= 35$$

Q.5. Compute the line integral $\int_C (y^2 dx - x^2 dy)$ about the triangle whose vertices are $(1,0)$, $(0,1)$ and $(-1,0)$.

$\Rightarrow$ Given;

Line integral $\int_C (y^2 dx - x^2 dy)$

On AB



$y = 0$

$dy = 0$

x varies from -1 to 1

$$\int_{-1}^{1} (0\,dx - x^2 0) = 0$$

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Page No.:
Date: ___/___/___

On BC;

Eqn of line;

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$$

$$y - 0 = \frac{-0+1}{-1+0}(x-1)$$

$$y = 1 - x$$

$$dy = -dx$$

x varies from 1 to 0

$$\int_1^0 (1-x)^2 dx - x^2(-dx)$$

$$\int_1^0 (1+x^2-2x)\,dx + x^2\,dx$$

$$= \int_1^0 (1+2x^2-2x)\,dx$$

$$= \left[x + \frac{2}{3}x^3 - x^2\right]_1^0$$

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$$= [0] - [1 + 2/3 - 1]$$

$$= -2/3$$

On (A);

Eqn of line;

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$$

$$y - 1 = \frac{0-1}{-1-0}(x - 0)$$

$$y - 1 = x$$

$$y = 1 + x$$

$$dy = dx$$

x varies from 0 to -1

$$\int_0^{-1} (1+x)^2 dx - x^2 dx$$

$$\int_0^{-1} (1 + x^2 + 2x) - x^2 dx$$

$$= \int_0^{-1} (1 + 2x) dx$$

$$= [x + x^2]_0^{-1}$$

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$= [-1+1]$

$= 0$

Total $= \int_{AB} () + \int_{BC} () + \int_{CA} ()$

$= 0 - 2/3 + 0$

$= -2/3$

Q.6. If $A = (3x^2+6y)\,I - 14yz\,J + 20xz^2\,K$, evaluate $\int A \cdot dR$ from $(0,0,0)$ to $(1,1,1)$ along the path $x = t,\ y = t^2,\ z = t^3$.

$\Rightarrow$ Given;

$A = (3x^2+6y)\,I - 14yz\,J + 20xz^2\,K$

$\int A \cdot dR = \int (3x^2+6y)\,dx - 14yz\,dy + 20xz^2\,dz$

along the path

$$\left.\begin{aligned} x &= t \\ dx &= dt \\ y &= t^2 \\ dy &= 2t\,dt \\ z &= t^3 \\ dz &= 3t^2\,dt \end{aligned}\right\} \quad t \text{ varies from } 0 \text{ to } 1$$

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Page No.:
Date: ___/___/___

$$= \int_0^1 (3t^2 + 6t^2)\,dt - 14t^2 \cdot t^3 \cdot 2t\,dt + 20t(t^3)^2 \cdot 3t^2\,dt$$

$$= \int_0^1 (3t^2 + 6t^2 - 28t^6 + 60t^9)\,dt$$

$$= \int_0^1 (9t^2 - 28t^6 + 60t^9)\,dt$$

$$= \left[\frac{9t^3}{3} - \frac{28t^7}{7} + \frac{60t^{10}}{10}\right]_0^1$$

$$= 3 - 4 + 6$$

$$= 5$$

Q-7. Evaluate $\int_C (xy + z^2)\,dx$, where, C is the arc of helix $x = \cos t$, $y = \sin t$, $z = t$, which joins the point $(1,0,0)$ and $(-1, 0, \pi)$.

=> Given

$$\int_C (xy + z^2)\,ds$$

arc of helix

$x = \cos t$ $\quad$ $y = \sin t$ $\quad$ $z = t$

$dx = -\sin t\,dt$ $\quad$ $dy = \cos t\,dt$ $\quad$ $dz = dt$

Akash

Page No.: _____
Date: ___/___/___

t varie from 0 to π

$$\int_0^{\pi} (\cos t \sin t + t^2) - \sin t \, dt$$

$$-\left[\int_0^{\pi} \sin^2 t \cos t \, dt + \int_0^{\pi} t^2 \sin t \, dt\right]$$

$$-\left[\sin^2 t \int_0^{\pi} \cos t \, dt - \int \frac{d}{dt} \sin^2 t \int \cos t \, dt + t^2 \int \sin t \, dt - \int \frac{d}{dt} t^2 \int \sin t \, dt\right]$$

$$-\left[\sin^2 t \left[\sin t\right]_0^{\pi} - 2\int \sin t \cos t \sin t \, dt\right]$$

$$\left[-t^2 \cos t\right]_0^{\pi} + 2\int t \cos t \, dt\Big]$$

Solve it.

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Q.8. Find the total work done in moving a particle in a force field given by $F = 3xy\,I - 5x\,J + 10x\,K$ along the curve the point $(0,0,0)$ to the point to the point $(2,1,1)$ along the curve $x = 2t^2, y = t, z = t^3$.

$\Rightarrow$ Given;

$$F = 3xy\,I - 5x\,J + 10x\,K$$

$$\int_C F \cdot dR = \int_C 3xy\,dx - 5x\,dy + 10x\,dz$$

along the curve

$x = 2t^2$

$dx = 4t\,dt$

$y = t$

$dy = dt$

$z = t^3$

$dz = 3t^2\,dt$

t varies from 0 to 1

$$\int_0^1 3(2t^2)(t)\,4t\,dt - 5(2t^2)\,dt + 10(2t^2)\,3t^2\,dt$$

$$= \int_0^1 24t^3\,dt - 10t^2 + 60t^4\,dt$$

$$= \left[6t^4 - \frac{10}{3}t^3 + 12t^5\right]_0^1$$

$$= 6 - \frac{10}{3} + 12 = \frac{54-10}{3} = 44/3$$

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